Do Now · retrieval

Lesson 056 · Tree Traversals: Post-order and Breadth-first

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a leaf node?

Q2

What is BFS's underlying structure?

Q3

What order does postorder use?

Q4

A binary tree has root 10, children 5 and 15, and node 5 has right child 7 only. Trace its postorder traversal, using left subtree, right subtree, root.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Post-order visits both child subtrees before its parent. Traverse left subtree, right subtree, then root.

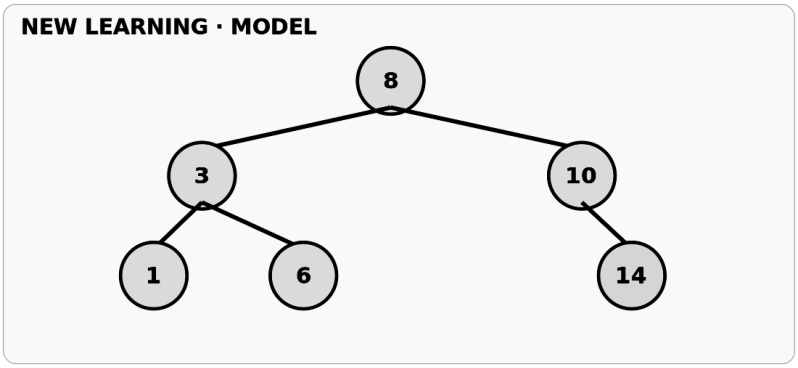
Core idea 2

Model — Breadth-first visits by levels, left-to-right with a queue that stores the next vertices.

Core idea 3

Evaluate — Traversal result depends on the precise tree and stated child order. BFS queue changes must be shown to justify an answer.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

postorder — Left-right-root tree traversal.

BFS — Breadth-first level exploration.

queue — FIFO worklist.

recursive — Process invoking itself on children.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

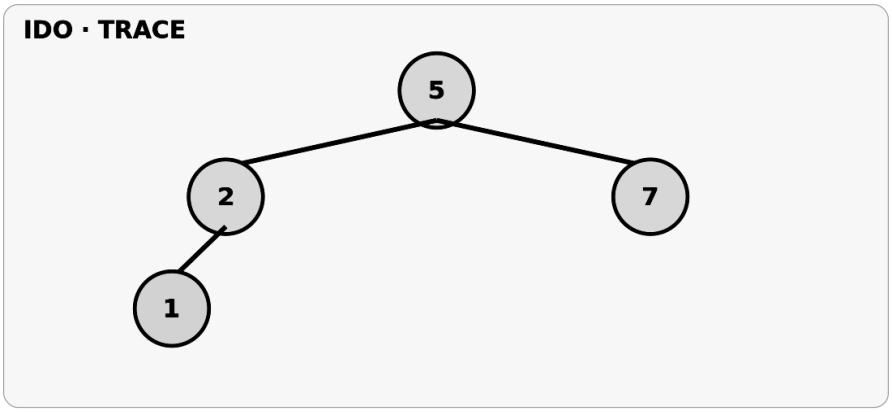
Problem

Give postorder for tree root5, left2, right7; node2 has left1.

Follow the worked method

1. Explain — PostOrder(5) waits for PostOrder(2), which prints 1 then 2. The right call prints 7; only then is 5 printed: 1,2,7,5. null returns without printing. Node supplies Key, Left and Right.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain
Order 9,4,12,6. Visit by depth, using FIFO.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Root8, children3 and10, with 3 children1 and6, 10 right child14. Give postorder. [6]

Question 2

Give breadth-first order for same tree, and first two queue changes. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why postorder is useful for deleting tree structures but BFS for shortest unweighted-edge distance. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Breadth-first tree traversal uses a stack to process deepest nodes first.

Use key vocabulary from memory

1. Explain postorder in your own words.

2. Explain BFS in your own words.

3. Explain queue in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.